A Planimetric Location Method for Laser Footprints of the Chinese Gaofen-7 Satellite Using Laser Spot Center Detection and Image Matching to Stereo Image Product

Huan Xie, *Senior Member, IEEE*, Binbin Li, Xiaohua Tong, *Senior Member, IEEE*, Xinwei Zhang, Tao He, Jun Dai, Genghua Huang, Zhijie Zhang, and Shijie Liu, *Member, IEEE*

***Abstract*— Satellite stereo mapping, together with laser altimetry, can be used to obtain three-dimensional geospatial information. Spaceborne laser altimeter can provide high-accuracy elevation information; however, due to the lack of detailed intensity information, its planimetric accuracy is usually worse than the ranging accuracy. The Chinese Gaofen-7 (GF-7) satellite, which was designed for civilian mapping application, was launched on November 3, 2019. The GF-7 satellite's main payloads are a laser altimeter system (with footprint camera) and a dual-linear charge-coupled device (CCD) mapping camera. According to the pixel coordinate of the laser footprint in the stereo image, the laser altimeter together with the footprint camera can provide planimetric geodetic coordinates for the control points of a higher accuracy than the other traditional satellite laser altimeters, and represents a new technology for satellite mapping. In this article, a laser footprint planimetric location method for the GF-7 satellite is proposed. The method is designed based on the main payload characteristics of GF-7 and the working modes of the laser altimeter by the combined use of subpixel phase correlation image matching and four types of laser spot center detection methods. The planimetric positioning accuracies of the laser spots in urban, suburban, farmland, forest, mountainous, and ice sheet areas were also analyzed. The experimental results show that the accuracy of planimetric location relative to stereo image for the laser footprint is 0.3–1.0 m (except for ice sheets ∼12 m) when the footprint camera works under the synchronous mode, and 0.2–0.4 m when the footprint camera works under asynchronous mode (AM).**

***Index Terms*— Footprint camera, Gaofen-7 (GF-7), laser altimetry, laser footprint positioning, stereo mapping.**

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Huan Xie is with the College of Surveying and Geo-Informatics, and Shanghai Key Laboratory for Planetary Mapping and Remote Sensing for Deep Space Exploration, Tongji University, Shanghai 200092, China and also with the Shanghai Institute of Intelligent Science and Technology, Tongji University, Shanghai 200092, China (e-mail: huanxie@tongji.edu.cn).

Binbin Li, Xiaohua Tong, and Shijie Liu are with the College of Surveying and Geo-Informatics, and Shanghai Key Laboratory for Planetary Mapping and Remote Sensing for Deep Space Exploration, Tongji University, Shanghai 200092, China (e-mail: libinbin@tongji.edu.cn; xhtong@tongji.edu.cn; liusjtj@tongji.edu.cn).

Xinwei Zhang, Tao He, and Jun Dai are with the Beijing Institute of Spacecraft System Engineering, China Academy of Space Technology, Beijing 100094, China (e-mail: 8619788@qq.com; zd09ht@163.com; daijunyu@qq.com;).

Genghua Huang is with the Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai 200083, China (e-mail: genghuah@mail.sitp.ac.cn).

Zhijie Zhang is with the School of Surveying and Land Information Engineering, Henan Polytechnic University, Jiaozuo, Henan 454000, China (e-mail: zhangzj2017@hotmail.com).

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I. Introduction

With the emergence and development of high-resolution satellite stereo imaging technology, great progress has been made in the application of these high-resolution remote sensing images in mapping applications such as digital surface model (DSM) generation [1]–[3] and national topographic mapping [4], [5]. However, the vertical accuracy of optical stereoscopic mapping is usually worse than the planimetric accuracy, due to the influence of the base-to-height ratio, the platform stability, and other factors, and it can be especially difficult to extract accurate elevation observations in weak texture areas [5]–[8]. To overcome this problem, satellite laser altimeters, which can provide high-accuracy ranging data, have been widely used in satellite photogrammetry [9], [10].

In deep space exploration, it is necessary to combine stereo imagery and laser altimeter data in order to extract high-precision elevation observations (not an absolute geolocation) of the surface of the Moon, Mars, and other planets, since there is a lack of ground control points and the surface textures are weak. A lot of work has been done in this regard, such as the integration of Clementine images and Clementine laser altimeter data [11]; the registration of Chang'E-1 images and laser altimeter data for accurate lunar topographic modeling [12], [13]; combined processing between Mars Orbiter Camera (MOC) imagery and Mars Orbiter Laser Altimeter (MOLA) data to correct the mis-registration and accurately determine

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ground positions [14]; and topographic mapping (at a 25-m contour interval) from high-resolution stereo camera (HRSC) and MOLA data to measure the morphometric characteristics of the structure in the morphologic features of the dome in the Tithonium Chasma of Mars [15].

In terms of Earth observation, the Shuttle Laser Altimeter 1 (SLA-01) was carried on-board the Space Shuttle Endeavor STS-72 mission and the SLA-02 was carried on-board the Space Shuttle Discovery STS-85 mission, which were launched in January 1996 and August 1997, respectively. SLA 01/02 were built from spare components of the MOLA instrument, and they captured the vertical distribution of the roughness elements within a 100-m diameter footprint on the Earth's surface [16], [17]. Apart from the above experimental payloads, the Geoscience Laser Altimeter System (GLAS) laser altimeter on-board the Ice Cloud and Land Elevation Satellite (ICESat), which was launched in 2003, is the first satellite-borne laser altimeter system in the world [18], [19]. Its primary objectives were focused on polar ice sheet mass balance, and it has also completed interdisciplinary applications in land topography, hydrology, vegetation canopy height estimation, cloud height estimation, and atmospheric aerosol distribution [19]. The follow-up ICESat-2 satellite was launched in September 2018, and ICESat-2 data were first released to the public in May 2019 [20]. ICESat-2 is the first space-borne platform with a photon counting laser altimeter. The photon counting laser altimeter can emit three pairs of laser beams (532 nm wavelength) at a laser repetition rate of 10 kHz. The multiple beam pairs can provide improved spatial coverage and the dense spatial sampling eliminates along-track measurement gaps compared to ICESat [21].

In China, the Ziyuan-3 (ZY-3) satellites, the first of which was launched on January 9, 2012, are the first high-resolution optical civil stereoscopic mapping satellites with integrated mapping and resource investigation functions [22], [23]. The ZY-3-02 satellite was launched on May 30, 2016, and carried China's first experimental laser altimetry payload for Earth observation, with the aim being to test the performance of the laser altimeter in the improvement of the elevation extraction accuracy of optical satellite imagery [24]. The GF-7 satellite was launched on November 3, 2019, and is a high-resolution stereo mapping satellite, with the main payloads being a laser altimeter (with footprint camera) and a dual-linear stereo mapping charge-coupled device (CCD) camera. The GF-7 satellite is aimed at achieving the application requirements of 1:10,000 stereo surveying and mapping [25], [26].

However, the spaceborne laser altimetry planimetric accuracy is generally lower than one-tenth of the ranging accuracy; for example, the ranging accuracy of GLAS altimetry data can reach 14 cm, while the planimetric accuracy is only about 10 m [19], [27]–[30]. Because of its lower planimetric accuracy, when there is obvious terrain undulation or surface object alternating with high and low, there will be deviation value or even mistake when using its vertical observation. Different strategies have been used to handle this problem. Some studies have fused stereo images and laser altimeter data to produce a digital elevation model (DEM); however, the geometric bias between the two data sets has not been carefully examined [31]. The second strategy is only using the vertical information to support the stereo block adjustment [32]. The third strategy is registering the stereo images to the laser altimeter data based on similarity transformation between the respective 3-D Cartesian coordinate systems or the 2-D image simplified point-to-line features of the two data sets in object space [7], [33]. The fourth strategy is using high-quality DEM/LiDAR data to simulate laser altimetry echo waveforms within an area that is close to the laser footprint (easting and northing ± 45.7 m). The planimetric geodetic coordinates of the laser footprint are then corrected by the planimetric offset that corresponds to a simulated echo waveform that has the maximum correlation coefficient with the echo waveform [34]. The fifth strategy is acquiring the accurate location of the laser footprint by airborne photography of the *in situ* laser footprint imaging detectors during a satellite overflight [35]. However, the first three strategies are greatly affected by the topographic distribution of the laser footprint. In regions with larger slopes, DEMs have a lower accuracy when they are generated by direct use of the 3-D information of the laser footprint. When using the second strategy to take the elevation observation of the laser footprint as an accurate vertical control point, the area where the laser footprint is located needs to be flat. The accuracy of the registration of laser footprints with stereo images using the third strategy depends on the distribution of the laser footprints in areas with strong texture characteristics and obvious elevation fluctuations. The fourth strategy is difficult to apply in areas without high-quality DEM or elevation data. The fifth strategy requires more detectors and longer realtime airborne photogrammetry with the increase of the investigation area.

Therefore, in order to improve the planimetric accuracy of the laser footprints for satellite optical stereo image adjustment and stereo mapping, GF-7's altimeter uses a footprint camera as an assistant. The footprint camera can record objects on the surface, and the laser footprint at the moment of laser emission, i.e., the initial point of transmission, which is a new technology for extracting the accurate planimetric geodetic coordinates of laser footprint centers in the stereo image base on pixel coordinate of the laser footprint in the stereo image. The image resolution and image texture complexity of the footprint image are very important for the implementation of this technology. A preliminary study has found that GF-7's footprint images [in asynchronous mode (AM)] are of a good quality, and the estimated error of the footprint centroid of the laser footprint centers between the footprint images is less than 0.7 pixels [36].

In the past research, the planimetric geodetic coordinates of the laser footprint were solved by a laser geometric model. In GF-7, the planimetric geodetic coordinates of laser footprint are determined by matching the laser footprint image with geometrically corrected stereo images. As there is no high precision attitude sensor directly mounted together with the laser altimeter, the method of planimetric location relative to stereo image for laser footprint is very important. The matching accuracy directly affects the geolocation accuracy of laser footprint, and the registration accuracy of stereo images and laser altimeter data for stereo mapping. In this process, the

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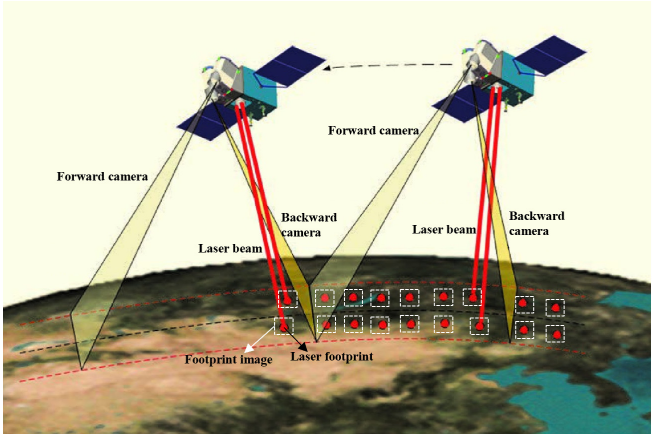


Fig. 1. GF-7's dual-linear stereo mapping CCD camera and laser altimeter detect the Earth's surface (modified from [37]). The scale in the figure is not strict, it's just a sketch.

technology for solving the planimetric geodetic coordinates of the stereo image is relatively mature. Therefore, in this article, we focus on the method of accurate planimetric location relative to stereo image for the laser footprint of the Chinese GF-7 satellite, simulated data sets generated from QuickBird and WorldView-3 images in urban, suburban, farmland, forest, mountainous, and ice sheet areas were used to validate the proposed method. We then analyze the planimetric positioning accuracy of the simulation results under the two different types of footprint camera mode. A description of the working modes of the footprint camera is given in Section II. The planimetric geodetic coordinate extraction method for the laser footprints is described in Section III, which includes the extraction of the laser footprint center's coordinate in the footprint image and coordinate transformation of the laser footprint from the footprint image to the stereo image. The simulation of the GF-7 footprint and stereo images is described in Section IV. The results and the experimental analysis are given in Section V, including analyses of the extraction of laser footprint center's coordinates in the footprint image, the image registration, the planimetric positioning accuracy, and geolocation accuracy of the stereo image. The conclusions of this study are given in Section VI.

II. Working Modes of the GF-7 Footprint Camera

As shown in Fig. 1, GF-7's dual-linear stereo mapping CCD camera and laser altimeter detect the Earth's surface on different optical axes. The dual-linear stereo mapping CCD camera does not image the laser emission beam, so that the camera is unable to locate the laser footprint in the stereo imagery directly. Moreover, the dual-linear stereo mapping CCD camera is rigidly connected with the accurate attitude sensors (star sensors et.al), while the laser altimeter system is not connected. The laser altimeter system needs to use the built-in footprint camera to locate the laser footprint coordinate in the stereo image, so as to obtain the high-accuracy laser emission direction and geodetic coordinate of laser footprint.

As shown in Fig. 2, in the light path of the laser altimeter, optical extraction modules are designed to extract part of the emitted laser. Among them, 98% of the laser energy passes through optical extraction modules and forms footprints on

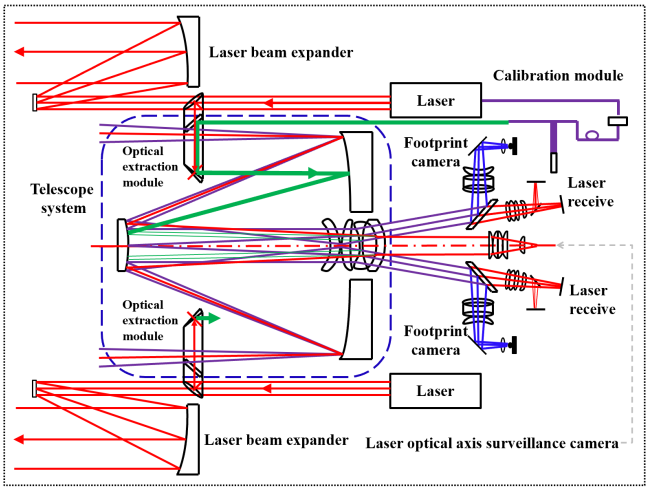


Fig. 2. Laser altimeter system design (modified from [38]).

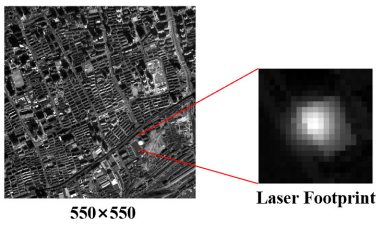


Fig. 3. Footprint image under SM of the footprint camera.

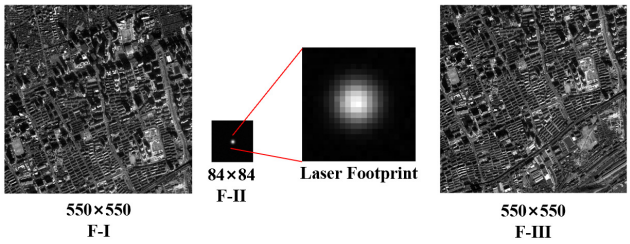


Fig. 4. Footprint image under AM of the footprint camera.

the ground after collimating with the laser beam expanders. The remaining 2% of the energy reaches the focal plane of the footprint camera (1%) and the optical axis monitoring camera through a telescope system, which is not transmitted through the atmosphere, and not reflected from the ground. Laser Optical Axis Surveillance Camera that is located at the center of the system simultaneously images two laser spots to monitor the relative positional relationship between the two beam lasers to monitor and analyze laser stability and its lifetime. Footprint camera can not only image the single beam of emitted laser, but also image the objects on the surface. Both of them are imaging on the same optical axis, the relative position of the laser footprint in the ground object can be determined.

According to different exposure sequences, GF-7's footprint camera has two different exposure imaging modes: synchronous mode (SM) and AM [39]. As shown in Fig. 3, SM allows effective imaging of the objects on the surface and the laser footprint at the moment of laser emission. As shown in Fig. 4, AM first allows effective imaging of the objects

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on the surface before the laser emission, to generate the first footprint image (F-I), and then allows effective imaging of the laser footprint at the moment of laser emission, to generate the second footprint image (F-II). In order to reduce the amount of data transmission, the F-II image only transmits the areas where the laser footprints are concentrated, but not other areas with the black background. Finally, it allows effective imaging of the objects on the surface after the laser emission, to generate the third footprint image (F-III). As a result of the short exposure time, the F-II image generally only contains the laser footprint, but may also contain dim objects on the surface when the footprint camera captures the strong reflected light of the objects in the camera's field of view [36].

III. Planimetric Geolocation Extraction for the Laser Footprints

The main processing steps for extracting high-accuracy planimetric geolocation for the laser footprints include the extraction of laser footprint center's coordinates in the footprint image, and coordinate transformation of the laser footprint centers from the footprint images to the stereo images.

A. Extraction of the Laser Footprint Center's Coordinate in the Footprint Image

The purpose of footprint detection is to find the laser spot in the footprint image. As shown in Fig. 3, if the laser footprint center was searched for in a whole footprint image, the processing would be inefficient and the detection accuracy would be poor. To avoid this situation, it is necessary to first narrow the search area, as follows.

1) A default value for the laser footprint center is used as the center to generate a square area, for which the length is four times the radius length of the laser footprint, which is named the "initial detection area." The value can be estimated by the relative position of the footprint camera and the laser beam embedded through the 0.1% reflection optical prism. The error of this value in the simulation experiments was about two pixels.
2) The location of the maximum gray value is detected from the initial detection area, and is regarded as the initial laser footprint center.
3) With the initial laser footprint center, a square area is selected, for which the length is two times the radius length of the laser footprint, which is the maximum distribution area of the laser footprint in the footprint image, and is named the "final detection area."

After obtaining the final detection area, the next step is to detect the laser spot center. As shown in Fig. 3, the laser footprint has an almost elliptical distribution, but it has a strong noise distribution, because the laser spot is affected by sensor noise (see Section IV) and the surface image. In order to better realize the extraction of the laser spot center, we used four commonly used laser spot detection methods to analyze the laser footprints, as described in Section V-C. These methods were the gray centroid method [40], Gaussian fitting [41], Gabor fitting [41], and ellipse fitting [42]. For the ellipse fitting, image binarization [43] and image edge detection [45] were the processing steps. The last three methods can automatically adjust the parameters according to the gray value distribution of the image to find the best fitting results.

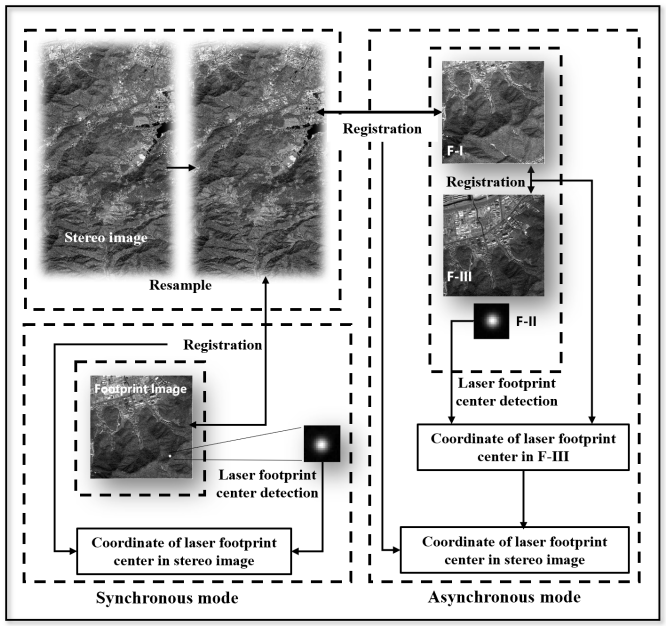


Fig. 5. Flowchart of coordinate transformation of the laser footprint center from footprint image to stereo image.

B. Coordinate Transformation of the Laser Footprint From Footprint Image to Stereo Image

As shown in Fig. 5, the process of coordinate extraction of the laser spot center in the stereo image is slightly different under the different working modes of the footprint camera.

Under SM, at first, the stereo image is resampled to the same resolution as the footprint image. The resampled stereo image is then registered with the footprint image to obtain the matching pairs between the two images. These matching pairs are then used to obtain the transformation relationship between the two images. The transformation relationship is the second-order polynomial model [45]. Finally, the coordinate transformation of the laser footprint center from the footprint image to the stereo image can be achieved based on the transformation relationship.

Under AM, at first, the F-II image is registered with the F-III image to obtain the matching pairs between the two images. These matching pairs are then used to obtain the transformation relationship between the two images. Since the imaging time interval between F-I and F-III is short (the time interval will be from 0.025 to 0.1 s because the laser emission frequency is 3 Hz and in order to ensure enough overlap between images (60%–90%)), the transformation relationship between the two images can be assumed to be approximately linear. Therefore, the coordinates of the laser spot center in F-III ($X_{\mathrm{F-III}}$, $Y_{\mathrm{F-III}}$) can be obtained according to the following formulas:

$$X_{\mathrm{F-III}} = X_{\mathrm{F-II}} + \Delta X \cdot [(T_{\mathrm{F-III}} - T_{\mathrm{F-II}})/(T_{\mathrm{F-III}} - T_{\mathrm{F-I}})] \quad (1)$$

$$Y_{\mathrm{F-III}} = Y_{\mathrm{F-II}} + \Delta Y \cdot [(T_{\mathrm{F-III}} - T_{\mathrm{F-II}})/(T_{\mathrm{F-III}} - T_{\mathrm{F-I}})] \quad (2)$$

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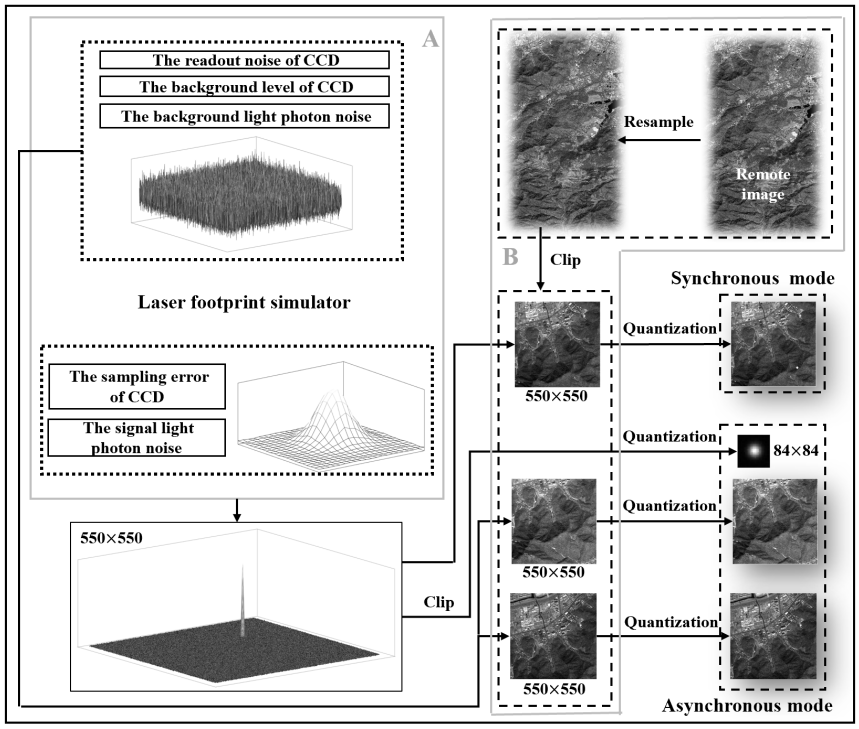


Fig. 6. Flowchart of stereo image and footprint image simulation.

where ΔX and ΔY are the translation values between F-I and F-III, $(X_{\mathrm{F-II}}, Y_{\mathrm{F-II}})$ represent the coordinates of the laser spot center in F-II, and $T_{\mathrm{F-I}}$, $T_{\mathrm{F-II}}$, and $T_{\mathrm{F-III}}$ are the imaging times of F-I, F-II, and F-III, respectively. After obtaining the coordinates of the laser spot center in F-III, the same process steps as those used for SM are used for the extraction of the coordinates of the laser spot center in the stereo image.

The image matching method used in this study was a subpixel phase correlation method, which can directly obtain accurate and reliable subpixel estimations in the Fourier domain [46], [47].

IV. Simulation of the GF-7 Footprint and Stereo Images

As the GF-7 satellite is still in the stage of Cal/Val, data have not been officially released, so we simulated the stereo images by resampling high-resolution remote sensing imagery to the same resolution as the GF-7 stereo imagery. The simulation process for the footprint image consisted of two main parts: 1) simulation of the laser footprint image; and 2) simulation of the Earth surface image in the footprint camera's field of view. The process of footprint image simulation is shown in Fig. 6.

A. Simulation of the Laser Footprint Image

As shown in area A of Fig. 6, the simulation process [48], [49] for the laser footprint image includes simulation of the readout noise of the CCD, the background light photon noise, the background level of the CCD, the sampling error of the CCD, and the signal light photon noise. Among these sources of noise, the first three terms, which are considered as being the first-class error source, are distributed over the whole of the CCD, and the last two terms, which are considered as the second-class error source, are only found in the area of the laser footprint [48]. The readout noise of the CCD is Gaussian distribution noise, for which the mean is 0; the signal light photon noise and the background light photon noise are Poisson distribution noise, for which the variance is the average signal photon number (or gray value) of a pixel in the laser footprint distribution area; the background level of the CCD is a constant value; and the sampling error of the CCD is a Gaussian distribution, which is directly correlated with the laser footprint size in a sub-aperture [48], [49]. To simulate these noise types, some of the parameters were assumed (see Section V-A), including the average value of the background level of the CCD, the profile (the peak, variance, and centroid coordinates) of the average photon density in the CCD, and the variance of the readout noise of the CCD and the background light photon noise. The footprint camera of GF-7 was imaged by a complementary metal–oxide–semiconductor (CMOS) sensor. The distributions of the main noise sources for the CMOS sensor are the same or similar to those for the CCD sensor [50], and its antinoise performance is generally slightly lower than that of the CCD sensor. Therefore, the simulated footprint image can be considered as having only a small difference in noise.

B. Simulation of the Earth Surface Image in the Footprint Camera's Field of View

The simulation process for the Earth surface image in the footprint camera's field of view is different under the different working modes of the footprint camera, as shown in area B of Fig. 6.

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TABLE I
PARAMETERS OF THE SIMULATION PROCESS

Parameters	Value
The scale size of the footprint image	550 × 550 pixels (or 84 × 84 pixels)
The instantaneous field of view (IFOV) of the footprint image	6.4 μrad
The resolution of the footprint image	3.2 m/pixel
The resolution of the stereo image	0.8 m/pixel
The digitization of the footprint image	10 bits
The digitization of the stereo image	11 bits
The planimetric offset of the left vertex coordinates of the footprint image relative to the laser footprint center coordinates	(391.625, 391.625) pixels
The peak of the average photon density in the CCD	2457 gray values
The variance of the average photon density in the CCD	2.5 pixels
The centroid coordinates of the average photon density in the CCD	(391.625, 391.625) pixels
The average value of the background level of the CCD	50 gray values
The variance of the readout noise of the CCD	5 gray values
The variance of the background light photon noise	100 gray values
The orbit altitude of the satellite	500 km
The laser emission frequency	3 Hz
The number of beams	2
The inter-beam angle	0.7 degrees

Under SM, there is an Earth surface image in the footprint camera's field of view that needs to be extracted by clipping the resampled high-resolution remote sensing image (the resolution of which is the same as that of the footprint image). The clipping process is based on the size of the footprint image and the left vertex coordinates of the footprint image. The coordinates (X_{left}, Y_{left}) can be computed according to the planimetric offset (ΔX_{offset}, ΔY_{offset}) relative to the laser footprint center coordinates (X_{laser}, Y_{laser}) in the resampled high-resolution remote sensing image, as shown in the following formulas:

$$X_{\text{left}} = X_{\text{laser}} + \Delta X_{\text{offset}} \tag{3}$$

$$Y_{\text{left}} = Y_{\text{laser}} + \Delta Y_{\text{offset}}. \tag{4}$$

Under AM, there are two Earth surface images in the footprint camera's field of view that need to be extracted, which are represented by the Earth surface images of F-I and F-III, respectively. The extraction process for each image is roughly the same as that under SM, except for the computation of the left vertex coordinates of each image. The computation process is shown in the following formulas:

$$X_{\text{left-I}} = X_{\text{laser}} + V_X \cdot \Delta t_{\text{I-II}} \tag{5}$$

$$Y_{\text{left-I}} = Y_{\text{laser}} + V_Y \cdot \Delta t_{\text{I-II}} \tag{6}$$

$$X_{\text{left-III}} = X_{\text{laser}} + V_X \cdot \Delta t_{\text{II-III}} \tag{7}$$

$$Y_{\text{left-III}} = Y_{\text{laser}} + V_Y \cdot \Delta t_{\text{II-III}} \tag{8}$$

where ($X_{\text{left-I}}$, $Y_{\text{left-I}}$) and ($X_{\text{left-III}}$, $Y_{\text{left-III}}$) are the left vertex coordinates of F-I and F-III, respectively. $\Delta t_{\text{I-II}}$ is the time difference between $T_{\text{F-II}}$ and $T_{\text{F-III}}$, and $\Delta t_{\text{II-III}}$ is the time difference between T_{FIML} and T_{FIMT}. (V_X, V_Y) is the velocity of the satellite in the *XY* direction. The *X* and *Y* directions of the satellite velocity coordinate system are consistent with the *X* and *Y* directions of the remote sensing image projection coordinate system. The *Z* direction of the satellite velocity coordinates system is consistent with that perpendicular to the *XY* plane of the remote sensing image. To ensure enough overlap between images (60%–90%), the imaging interval between F-I and F-III is very short (from 0.025 to 0.1 s). It is assumed that the satellite height (*Z* direction of remote sensing image) remains unchanged in F-I and F-III. According to the relevant research [51], the errors associated with the resampling of the images for the simulation study are relatively small, and we ignore its impact.

The values of $\Delta t_{\text{I-II}}$ and $\Delta t_{\text{II-III}}$ are set according to the exposure interval between the footprint images. Different exposure intervals will result in different overlapping areas between F-I and F-III. The larger the overlap area, the higher the accuracy of the image registration. According to the different exposure intervals, there will be about 60–90% image overlap. For example, an overlap of about 90% was reported in a previous study [36]. In the simulation experiments, we set these two values to 0.05 s, according to an image overlap of about 60%, as shown in Table I.

C. Footprint Image Generation From the Laser Footprint Image and Earth Surface Image

After completing the processing steps described in Sections IV-A and IV-B, the footprint image under SM can be obtained by superimposing its Earth surface image with the laser footprint image, F-I and F-III under AM can be obtained by superimposing their Earth surface images with the first-class error source, and F-II under AM can be obtained by clipping the laser footprint image.

Since the quantization digits of the footprint imagery were different from those of the remote sensing imagery, it was necessary to re-quantify the footprint imagery. Quantization processing is based on normalization from a zero value to each footprint image's maximum value, which is then multiplied by the maximum quantization range of the footprint image.

V. RESULTS AND ANALYSIS

A. Simulation Parameters

The parameters used in the simulation of the footprint images are listed in Table I, including the size and resolution of the footprint images, the satellite orbit height, and the related

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TABLE II
DETAILS OF THE QUICKBIRD AND WORLDVIEW-3 IMAGES

Area	High-resolution remote image	Acquisition date	Image size (km)	Pixel (m)	Description
Shanghai (urban)	QuickBird	2004/02/15	18.538 × 18.226	(~0.68, ~0.64)	1st image
Chongming (countryside)	QuickBird	2010/11/12	17.140 × 15.998	(~0.56, ~0.45)	2nd image
Urban	QuickBird	2004/02/15	4 × 8	(~0.68, ~0.64)	Subarea of the 1st image
Suburban	QuickBird	2010/11/12	4 × 8	(~0.56, ~0.45)	Subarea of the 2nd image
Farmland	QuickBird	2010/11/12	4 × 8	(~0.56, ~0.45)	Subarea of the 2nd image
Forest	QuickBird	2004/03/04	4 × 8	(0.6,0.6)	3rd image
Mountainous	QuickBird	2008/09/01	~7.11 × ~3.72	(0.6,0.6)	4th image
Ice sheet	WorldView-3	2017/10/04	~5.29 × ~5.32	(~0.35, ~0.35)	5th image

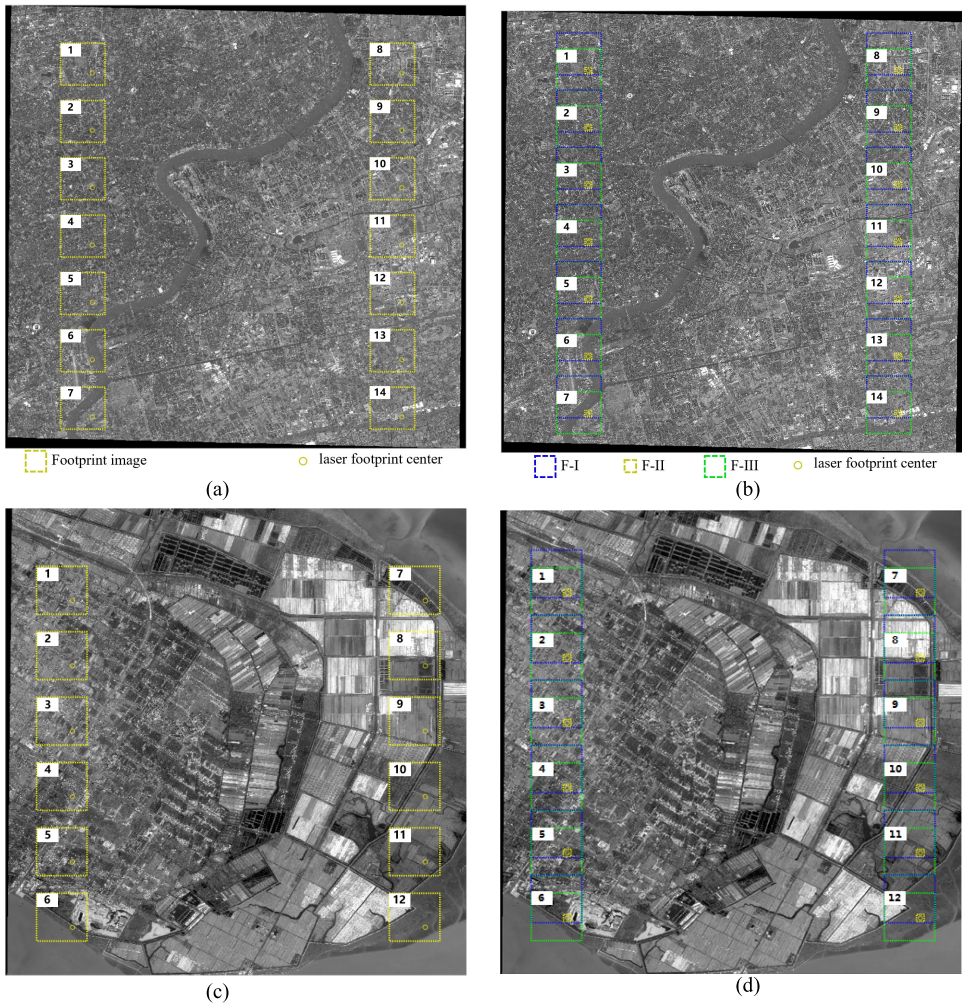


Fig. 7. Simulated footprint images in the areas of Shanghai (a) SM, (b) AM, and Chongming (c) SM, (d) AM, considering the interbeam angle.

simulation parameters for the laser footprints. The resolutions of the stereo images are ≤0.8 m (generally from 0.65 to 0.8 m), and the resolutions of the footprint images are about 3.2 m [26], [39]. In this study, resolutions of 0.8 m and 3.2 m were used to simulate the stereo images and the footprint images, respectively.

B. Simulation Data Sets

In this study, five high-resolution remote sensing images (QuickBird and WorldView-3) were used to simulate the footprint and stereo images. The details of these images are provided in Table II.

In Table II, only the image sizes of the first and second high-resolution remote sensing images are large enough to simulate the distribution of the two-beam footprint images of GF-7. These two images' simulation results for the footprint images are shown in Fig. 7. Moreover, we selected three different subareas (urban, suburban, and farmland) from the first and second high-resolution remote sensing images, and then combined them with the other three high-resolution remote sensing images (forest, mountainous, and ice sheet areas) to further analyze the performance of the laser footprint planimetric positioning technology in the six types of different land surfaces. The three selected subareas were of a similar

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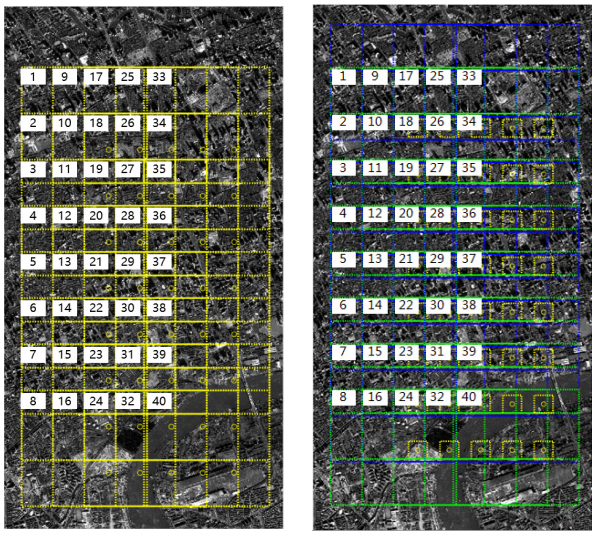


Fig. 8. Simulated footprint images in the urban area (a) SM, (b) AM.

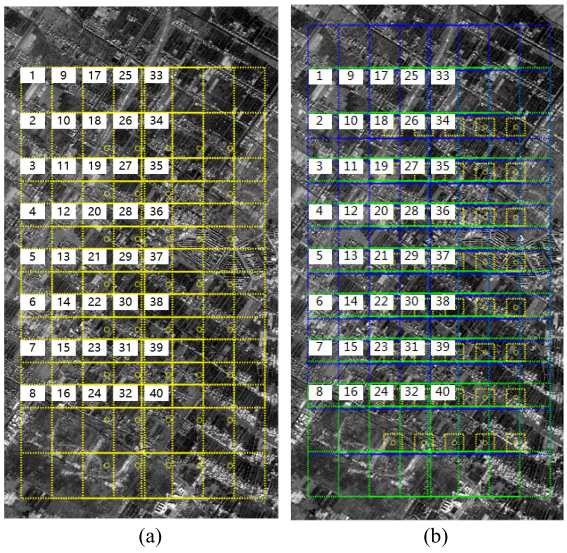


Fig. 9. Simulated footprint images in the suburban area (a) SM, (b) AM.

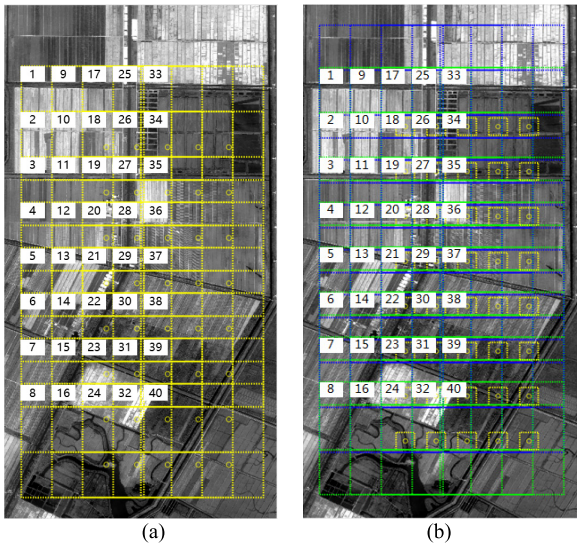


Fig. 10. Simulated footprint images in the farmland area (a) SM, (b) AM1.

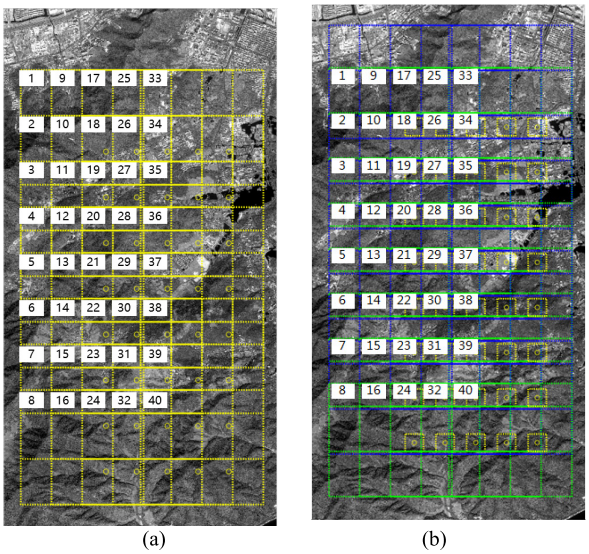


Fig. 11. Simulated footprint images in the forest area (a) SM, (b) AM.

image size to the other three high-resolution remote sensing images, and each of these six images was used to simulate 40 SM footprint images and 40 groups of AM footprint images. Each group of AM footprint images included the F-I, F-II, and F-III images. These footprint images only consider the case of the satellite single-beam laser emitting mono pulses, because of the limitation of the image size. The results for the simulated footprint images are shown in Figs. 8–13.

C. Analysis of the Results of the Laser Spot Detection in the Footprint Image

After obtaining the laser footprint extraction area according to the method described in Section III-A, the coordinates of the spot centers in the footprint images could be obtained by a laser spot detection method. We used four common detection methods to analyze the extraction accuracy of the laser spot centers: the gray centroid method, Gaussian fitting, Gabor fitting, and ellipse fitting. For each method, the spot center detection errors were obtained by calculating the distances between the coordinates of the spot center detection results and the coordinates of the simulated real laser footprint centers (shown in Figs. 7–13).

The spot center detection errors are shown in Figs. 14–16 (Fig. 14 for the errors in Shanghai and Chongming, and Figs. 15 and 16 for the errors in the SM and AM working modes for the six regions of different land-cover types).

From Figs. 14–16, it can be seen that the accuracy of the laser spot center detection under SM (error average is higher than 5 m) is lower than that under AM (error average is lower than 1 m). The main reason for this phenomenon is that the laser footprint image under SM is affected by surface reflection, while the F-II image under AM generally only contains the laser footprint, because of the short exposure time, as mentioned in Section II.

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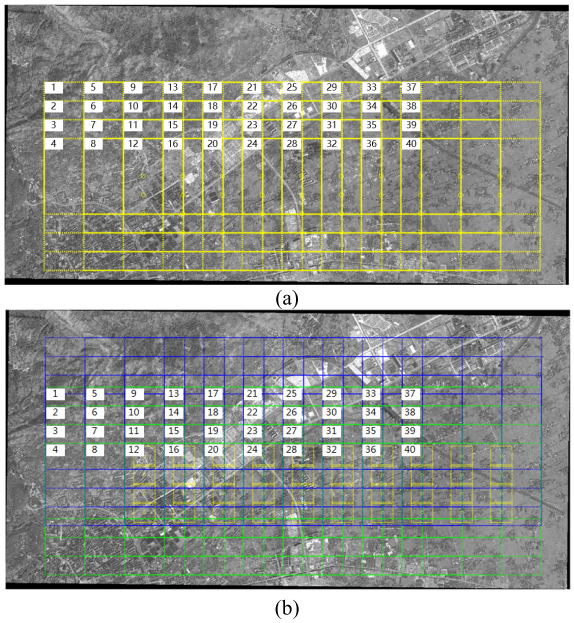


Fig. 12. Simulated footprint images in the mountainous area (a) SM, (b) AM.

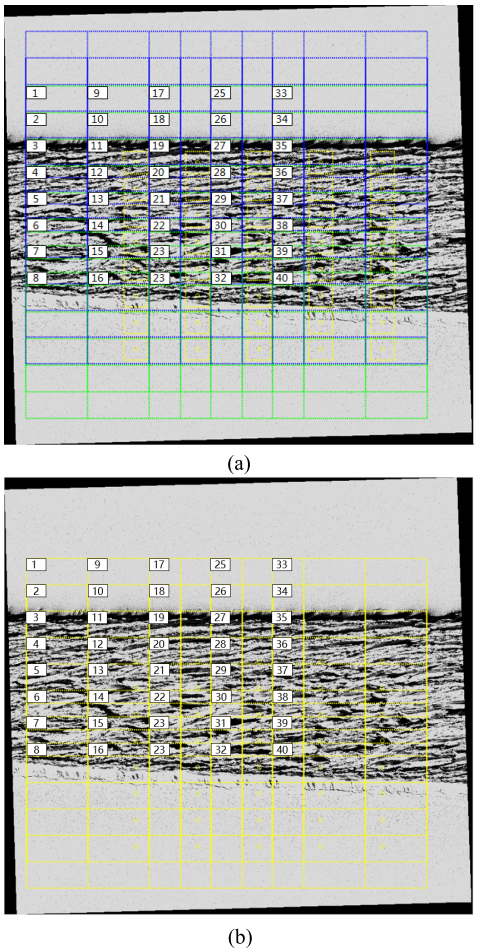


Fig. 13. Simulated footprint images in the ice sheet area (a) SM, (b) AM.

From the spot center detection errors, it can be seen that the ellipse fitting method is the best method for detecting the laser spot centers of the footprint images under SM, which

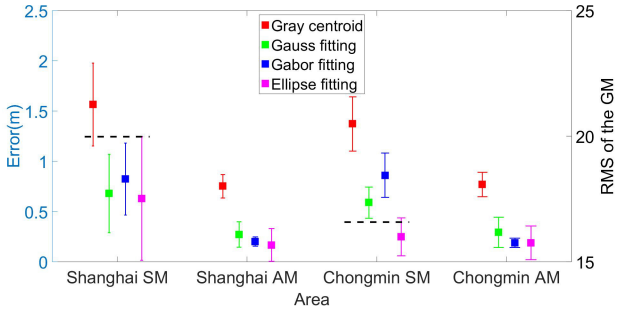


Fig. 14. Analysis of the laser spot center detection in the footprint images in the areas of Shanghai and Chongming.

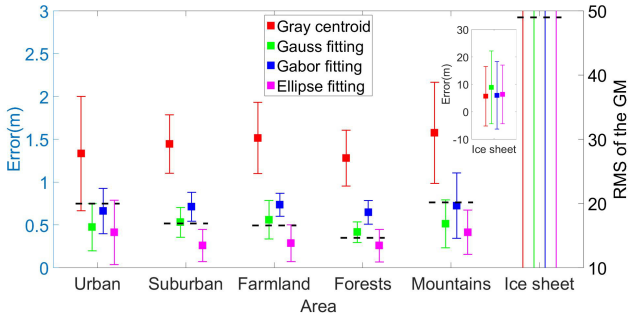


Fig. 15. Analysis of the laser spot center detection in the footprint images under SM in the six types of surface area.

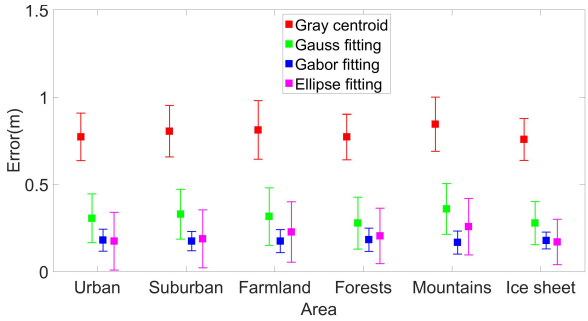


Fig. 16. Analysis of the laser spot center detection in the footprint images under AM in the six types of surface area.

may be because when the method detects a laser spot center, the influence of the surface reflections is mainly located in the spot edges, rather than the whole spot. Moreover, it is apparent that the accuracies of the laser spot center detection are clearly different in different areas. The accuracy of the laser spot center detection in the urban area of Shanghai is lower than that in the Chongming County area of Shanghai. In the six types of land cover, the accuracy of the laser spot center detection in the urban, mountainous, and ice sheet areas is lower than that in the suburban, farmland, and forest areas. To further analyze this phenomenon, we calculated the gradient median (GM) of the images to approximately characterize the reflectivity complexity of the images. The root-mean-square errors (RMSEs) of the footprint images' GM for the urban area of Shanghai and the Chongming County area of Shanghai are shown in Table III, and are marked with the dotted lines in

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TABLE III
RMSE OF THE GM IN THE SHANGHAI AND CHONGMING AREAS

Area	Footprint image number	RMSE of the GM (gray-value/pixel)
Shanghai (urban)	14	20.55
Chongming (countryside)	12	15.87

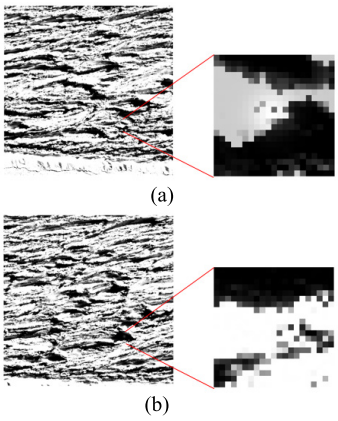


Fig. 17. Fourth and 36th footprint images in the ice sheet area. (a) Fourth footprint image. (b) Thirty-sixth footprint image.

the SM area of Fig. 14. The RMSEs of the footprint images' GM for the urban, suburban, farmland, forest, mountainous, and ice sheet areas are shown in Table IV, and are marked with the dotted lines in Fig. 15. From Figs. 14 and 15, it can be seen that the RMSE of the GM in the Chongming County area of Shanghai is lower than that in the urban area of Shanghai, and the RMSE of the GM in the suburban, farmland, and forest areas is lower than that in the urban, mountainous, and ice sheet areas. The higher the complexity of the surface reflectance, the lower the accuracy of the laser spot center detection. For the ice sheet area in particular, since it has high reflectivity complexity (the RMSE of the GM is 48.98 gray-value/pixel), the average laser footprint extraction error in the footprint image is more than 5 m, because the laser footprints are greatly weakened by the Earth's surface, as in the 4th and 36th footprint images in this area, which are shown in Fig. 17.

From Figs. 14 and 16, it can be seen that the Gaussian fitting, Gabor fitting, and ellipse fitting methods all show good detection performances (about 0–0.5 m error) for the laser spot center. Among the three methods, the Gabor fitting method is especially good for the laser spot center under AM, and the ellipse fitting method is especially good for the laser spot center under SM.

Under both SM and AM, the gray center of gravity method performs poorly in the detection of the laser footprint center because it has the worst ability to suppress noise among the four detection methods.

For the actual operation of satellite, the background noise will be the main factor affecting the accuracy of laser spot detection of footprint image, and the detection of laser spot center in footprint image under SM is more difficult than that under AM. Therefore, in order to analyze the anti-noise performance of four different methods for laser spot

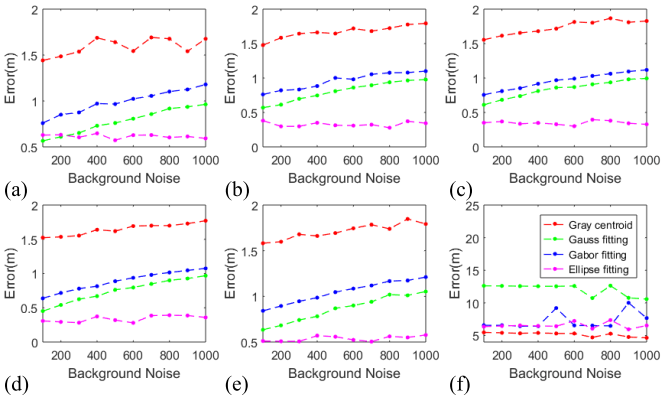


Fig. 18. Analysis of the laser spot center detection in the footprint images under AM in the six types of surface area with different background noise. (a)–(f) represent the analysis in urban, suburban, farmland, forests, mountains, and ice sheet, respectively. Each dot in sub-figure represents rms of laser spot center detection error of 40 footprint images that corresponding background noise.

center detection with higher background noise, the different background noises of the footprint images under SM were simulated in the six types of surface area. In the simulation experiment, the background noise of a footprint image starts with 100 gray values, and increases to 1000 gray values with step size of 100 gray values. For each type of surface area, 40 footprint images under SM were simulated under different gray values, which include 2400 footprint images in the six types of surface area. As shown in Fig. 18, the results show that, except for ice sheet, the anti-noise performance of the ellipse fitting method in other areas remains stable with the increase of background noise, while the anti-noise performance of the other three methods decreases with the increase of background noise. For ice sheet, due to the high reflectivity complexity described above plays a leading role, the laser spot detection errors of the four methods are relatively larger, more than 4 m.

D. Analysis of the Results of the Image Registration

In order to better analyze the planimetric positioning accuracy of the laser footprints in the different regions, we analyzed the registration accuracy between the footprint and stereo images in each region. Fig. 19 shows the image registration errors for the processing depicted in Fig. 7. Fig. 20 is a boxplot analysis of the registration errors for the six types of land cover.

From Figs. 19 and 20, it can be seen that the registration accuracy between the footprint images under AM (R-AM1) is generally higher than that between the footprint and stereo images under SM or AM (R-SM or R-AM2), which may be due to the sensors having the same quantization number and the same noise characteristics. In Fig. 20, there are obvious differences in the registration accuracy for the six types of Earth surface area. The registration accuracy for the farmland area is the worst, followed by the ice sheet, suburban, and forest areas, and the urban and mountainous areas show the highest accuracy. Moreover, in the areas with low reflectivity complexity (where the RMSE of the GM is about 14–16), the registration errors for the three

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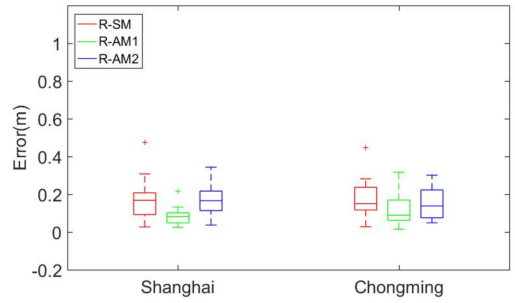


Fig. 19. Image registration errors in Shanghai and Chongming. (R-SM represents the registration error between the footprint images and stereo images under SM. R-AM1 represents the image registration error between the F-I and F-III images under AM. R-AM2 represents the registration error between the F-III images and the stereo images under AM).

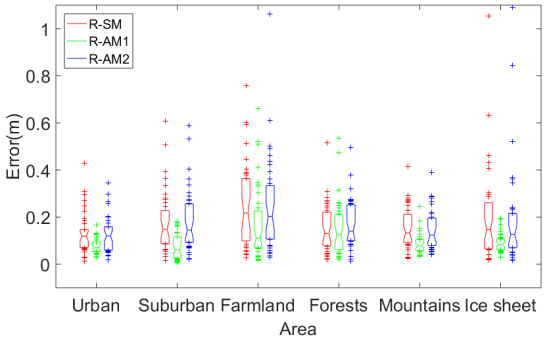


Fig. 20. Image registration errors in the different land-cover types. (R-SM represents the registration error between the footprint images and stereo images under SM. R-AM1 represents the image registration error between the F-I and F-III images under AM. R-AM2 represents the registration error between the F-III images and the stereo images under AM).

TABLE IV

RMSE OF THE GM IN THE URBAN, SUBURBAN, FARMLAND, FOREST, MOUNTAINOUS, AND ICE SHEET AREAS

Area	Footprint image number	RMSE of the GM (gray-value/pixel)
Urban	40	19.98
Suburban	40	16.89
Farmland	40	16.58
Forest	40	14.65
Mountainous	40	20.17
Ice sheet	40	48.98

TABLE V

FOOTPRINT PLANIMETRIC POSITIONING ERROR (RMSE) IN SHANGHAI AND CHONGMING

Area	**RMSE (m)**	
	SM	AM
Shanghai	0.9344	0.2417
Chongming	0.3572	0.3770

TABLE VI

FOOTPRINT PLANIMETRIC POSITIONING ERROR (RMSE) IN THE DIFFERENT LAND-COVER TYPES

Land cover	**RMSE (m)**	
	SM	AM
Urban	0.6003	0.2313
Suburban	0.3576	0.2816
Farmland	0.4452	0.4070
Forest	0.3470	0.3434
Mountainous	0.5102	0.2551
Ice sheet	12.3455	0.3526

types of images are significantly increased, and especially the registration error between the footprint images, as shown in the registration errors of the suburban, farmland, and forest areas in Fig. 20. It is in fact disadvantageous to implement the process of coordinate transformation of the laser footprint center from footprint image to stereo image under AM because the registration needs to be processed twice under AM and only once under SM.

E. Analysis of the Planimetric Positioning Accuracy

In Section V-C, we showed that the ellipse fitting method and Gabor fitting method have the best detection performance under SM and AM, respectively, so these two methods were used in the analysis of the planimetric positioning of the laser spot centers. The accuracy analysis of the planimetric positioning extraction results for the laser spot centers in the footprint images shown in Fig. 7 is provided in Table V, and the accuracy analysis of the planimetric positioning extraction results for the laser spot centers of the footprint images shown in Figs. 8–13 is provided in Table VI.

From Tables V and VI, it can be seen that the accuracy of the planimetric positioning extraction of the laser footprint image centers in AM is generally better than that in SM.

In the areas with low reflectivity complexity (where the GM is 14–16), such as Chongming and the suburban, farmland, and forest areas, the planimetric positioning accuracy of the footprint images under SM is close to that under AM, and the accuracy difference between the two modes is from about 0.36 (0.3470–0.3434 m) to 7.6 (0.3576–0.2816 m) cm. There are two reasons for this phenomenon. On the one hand, the detection accuracy for the laser footprint centers under SM in these areas is higher than that in the other areas (as analyzed in Section V-C). On the other hand, the image registration accuracy of the two modes in these areas is low, which means that it is disadvantageous to implement the process of coordinate transformation of the laser footprint centers from footprint images to stereo images under AM because the registration needs to be processed twice under AM and only once under SM (as analyzed in Section V-D).

In the areas with high complex reflectance (where the GM is ∼19–48), such as Shanghai and the urban, mountainous, and ice sheet areas, the planimetric positioning accuracy under AM is better than that under SM, and the accuracy difference between the two modes is from about 0.25 to 12 m. The reason for this phenomenon is that the detection accuracy for the laser footprint centers under SM is lower than that in the other areas (as analyzed in Section V-C), and the image registration accuracy of the two modes is high in these areas, especially between F-I and F-III under AM (as analyzed in Section V-D). As a special example, under SM, the positioning accuracy of the footprint centers is relatively poor in the ice sheet area,

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TABLE VII
ON-ORBIT TESTING RESULTS OF MAIN INFLUENCING FACTORS OF STEREO PLANIMETRIC POSITIONING ACCURACY OF STEREO IMAGE

Influencing factor	Error/Parameter	Design	On-orbit testing
Orbit	Orbit random error	0.1 m	0.1 m
	Orbit system error	0.1m	0.1m
	Output frequency	1Hz	1Hz
	Time synchronization accuracy	20μs	20μs
Attitude	Attitude random error	1.2″	0.3″
	Attitude system error	1″	1″
	Attitude stability	10^{-4}(°/s, 3σ)	6 10^{-5}(°/s, 3σ)
	Output frequency	8Hz	8Hz
Dual linear stereo mapping CCD camera	Pixel size	7μm	7μm
	Principal distance /mm	5520	Forward:5545.278 Backward:5541.507

at more than 12 m. This is mainly because the RMSE of the GM in this area reaches 48.98 gray-value/pixel, resulting in a huge footprint center detection error. In this kind of area, the positioning results for the footprint image centers should be used with caution, especially when accurate mapping is required.

Moreover, as mentioned in Section II, F-II under AM may also contain dim objects on the surface in the condition of objects with strong reflected light, which further reduces the detection accuracy of the laser footprint centers, thus reducing the advantage of AM over SM.

It is noteworthy that the accuracy of the laser footprint positioning technology studied in this article was established using stereo images. The actual positioning accuracy of this technology also needs to consider the planimetric error of the stereo image. Ignoring the impact of atmospheric error, as shown in Table VII, the relevant research simulation test shows that the planimetric positioning accuracy of stereo image can be better than 5 m without GCPs based on the orbit error, the attitude error, time synchronization accuracy, and dual linear stereo mapping CCD camera parameters of sate on orbit [38]. The stereo positioning accuracy can be further improved by GCPs. Relevant studies [52]–[54] have found that, in flat areas, such as urban and suburban areas, the corrected image accuracy is about 0.5–0.8 pixels, while in hilly and mountainous areas, such as rural and forest areas, the corrected image accuracy is about 1.1–3.2 pixels.

VI. CONCLUSION

In this article, we have proposed a geolocation method for the laser footprints of the Chinese Gaofen-7 (GF-7) satellite using stereo images and footprint images, which is a new approach that could improve the accuracy of the planimetric geolocation of control points for satellite optical stereo image adjustment and stereo mapping. By using a laser footprint simulation model to simulate several footprint images, we analyzed the accuracy of the planimetric geolocation extraction for laser spot centers in different land-cover types, with different working modes of GF-7 footprint camera. We found that the positioning accuracy of the laser footprint centers under SM was generally lower than that under AM. The experimental results showed that the accuracy of planimetric location relative to stereo image for the laser footprint was 0.3–1.0 m (except for the ice sheet areas, for which the accuracy of planimetric location relative to stereo image for the laser footprint was 12 m) when the footprint camera works under SM, and 0.2–0.4 m when the footprint camera works under AM. It was also found that the positioning accuracy of the laser footprint centers under SM was more affected by surface reflectivity. The higher the complexity of the surface reflectivity, the lower the accuracy of the laser footprint extraction in the footprint image. Moreover, among the different detection methods for laser footprint centers, the gray center of gravity method performed poorly for the footprint images because it has the worst ability to suppress noise. In addition, it was found that the performance of the ellipse fitting method was the best under SM, and the performance of the Gabor fitting method was the best under AM. In our future work, the influence of the slope on the actual positioning accuracy of the laser footprints extracted in a stereo mapping experiment with different terrain (e.g., flat land, hilly land, and mountainous areas) will be further analyzed.

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Huan Xie (Senior Member, IEEE) received the B.S. degree in surveying engineering, and the M.S. and the Ph. D. degrees in cartography and geoinfomation from Tongji University, Shanghai, China, in 2003, 2006, and 2009, respectively.

From 2007 to 2008, she was with the Institute of Photogrammetry and GeoInformation, Leibniz Universität, Hannover, Germany, funded by the China Scholarship Council, as a Visiting Scholar. Her research interests include satellite laser altimetry and hyperspectral remote sensing.

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Binbin Li received the M.S. degree in computer science and engineering from Shanghai Ocean University, Shanghai, China, in 2017. He is pursuing the Ph.D. degree in photogrammetry and remote sensing from the College of Surveying and Geo-Informatics, Tongji University, Shanghai, China.

His research interests include satellite laser altimetry and image processing for high-resolution satellite images.



Xiaohua Tong (Senior Member, IEEE) received the Ph.D. degree from Tongji University, Shanghai, China, in 1999.

From 2001 to 2003, he was a Post-Doctoral Researcher with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China. He was a Research Fellow with Hong Kong Polytechnic University, Hong Kong, in 2006, and a Visiting Scholar with the University of California, Santa Barbara, CA, USA, from 2008 to 2009. His research interests include photogrammetry and remote sensing, trust in spatial data, and image processing for high-resolution satellite images.



Xinwei Zhang is the Vice Project Manager of remote sensing satellite. His research interests include satellite overall design and remote sensing theory.



Tao He received the M.S. degree in astronautics from China Academy of Space Technology, Beijing, China, in 2016.

He is now an Engineer and a member of Remote Sensing Satellite Design Group. His research interests include satellite payload overall design and Lidar theory.



Jun Dai was born in 1981. He received the Ph.D. degree in optical engineering from Huazhong University of Science and Technology, Wuhan, China, in 2009.

His work focuses on mission analysis and design of optical remote-sensing satellite. He specializes in the space-borne optical payload such as camera and LIDAR. He is a Senior Engineer of China Academy of Space Technology (CAST), Beijing, China.



Genghua Huang received the Ph.D degree in physical electronics from the Chinese Academy of Science (CAS), Shanghai, China, in 2007.

He is a Senior Researcher with the Shanghai Institute of Technical Physics (SITP), CAS, and the Deputy Director of the designers in GF-7 laser altimeter system. His research interests in space-borne laser payloads, photon counting LIDAR system, and active remote sensing.



Zhijie Zhang received the Ph.D. degree in geomatics science and technology from Henan Polytechnic University, Jiaozuo, China, in 2020,

From 2017 to 2020, she was with the Tongji University, Shanghai, China, as a Visiting Graduate. She is a Post-Doctoral Researcher with the Center for Astro-geodynamics Research, astrometry and celestial mechanics, Shanghai Astronomical Observatory, CAS, Shanghai, China. She is under the joint supervision of Henan Polytechnic University and Tongji University. Her research interests include satellite laser altimetry theory and its applications.



Shijie Liu (Member, IEEE) received the B.S. in surveying engineering, and the M.S. and Ph.D. degrees in cartography and geoinfomation from Tongji University, Shanghai, China, in 2005, 2008, and 2012, respectively.

He is an Associate Professor of photogrammetry and remote sensing with the College of Surveying and Geoinformatics, Tongji University. His research interests include geometric exploitation of high-resolution remote sensing and its applications.

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